HR Analytics & Employee Performance

Challenge-Based Trainer Guide | Python Data Analysis | 2 Hours

Dataset note: synthetic records for training only. HR metrics should support fair, contextual review—not individual employment decisions.

1. Business scenario

A medium-sized company wants to understand why performance and satisfaction vary across departments. HR asks: Where are the strongest gaps? How do satisfaction, workload, training, salary, and absence relate to performance? What should be investigated next? Students work in teams and receive hints rather than a complete solution at the start.

2. Challenge outputs

• A short data-quality report (missing values, duplicates, invalid ranges, outliers).

• At least 4 department-level KPIs: headcount, average performance, satisfaction, salary, and absence.

• One Matplotlib chart, one Seaborn chart, and one interactive Plotly chart.

• A Streamlit dashboard with filters and five KPI cards.

• Three evidence-backed observations and two cautious, actionable recommendations.

3. Two-hour challenge schedule

Time	Challenge	Hint / checkpoint
0–10 min	Read the scenario; write 3 questions HR needs answered.	Use department, performance, satisfaction, salary, training, workload, and absence.
10–25 min	Import and profile the data.	Try shape, head, info, isna().sum(), duplicated().sum().
25–45 min	Clean and validate records.	Scores should be 1–5; salary must be positive; absence cannot be negative.
45–70 min	Build features and group-level metrics.	Use groupby().agg(); compute medians/percentiles with NumPy.
70–90 min	Choose charts that answer the questions.	Bar chart for department comparison; boxplot for spread; scatter for relationships.
90–115 min	Run and inspect the Streamlit dashboard.	Use filters to compare department, job level, status, and work mode.
115–120 min	Present one finding, one limitation, and one recommendation.	State the metric, comparison group, and what should be checked next.

4. Guided code checkpoints

Import and inspect

```
raw = pd.read_csv("hr_employee_raw.csv")
print(raw.shape)
print(raw.head())
print(raw.info())
print(raw.isna().sum())
print("Duplicates:", raw.duplicated().sum())
```

NumPy: use median and percentiles to summarize salary without relying only on the maximum.

```
salary = pd.to_numeric(raw["Monthly_Salary_EGP"], errors="coerce").dropna().to_numpy()
print(np.median(salary), np.percentile(salary, 90))
```

5. Cleaning hint (not a full answer key)

```
df = raw.copy()
df["Hire_Date"] = pd.to_datetime(df["Hire_Date"], errors="coerce")
df = df.drop_duplicates()
for col in ["Satisfaction_Score", "Performance_Score", "Workload_Score"]:
    df.loc[~df[col].between(1, 5), col] = np.nan
df.loc[df["Monthly_Salary_EGP"] <= 0, "Monthly_Salary_EGP"] = np.nan
df.loc[df["Absence_Days"] < 0, "Absence_Days"] = np.nan
df["Department"] = df["Department"].fillna("Unknown")
df["Monthly_Salary_EGP"] = df["Monthly_Salary_EGP"].fillna(
    df.groupby("Department")["Monthly_Salary_EGP"].transform("median"))
df["Satisfaction_Score"] = df["Satisfaction_Score"].fillna(df["Satisfaction_Score"].median())
df["Performance_Score"] = df["Performance_Score"].fillna(df["Performance_Score"].median())
```

6. Visualization and library mapping

• NumPy: median, percentile, clipping/validation checks, numerical arrays.

• Pandas: type conversion, missing values, duplicates, groupby, agg, correlation, derived columns.

• Matplotlib: static department performance bar chart.

• Seaborn: salary distribution boxplot or correlation heatmap.

• Plotly: interactive satisfaction-versus-performance scatter plot with hover details.

• Streamlit: sidebar filters, KPI cards, charts, data-quality panel, filtered-data download.

7. Discussion prompts and answer cues

Does satisfaction cause higher performance? No causal claim can be made from this cross-sectional synthetic dataset; it can only suggest an association.

Which department is best? Compare averages alongside headcount and score distributions; small groups and role mix may distort comparisons.

Does higher salary mean better performance? Not necessarily. Compare similar job levels and roles; department-level salary averages are confounded by job mix.

What can training tell us? Explore association with performance, but evaluate impact using pre/post measures or a controlled pilot.

What is a responsible recommendation? Investigate a group-level pattern, consult managers and employees, test an intervention, and monitor outcomes.

8. Dashboard run instructions

• Install: pip install -r requirements.txt

• Run: streamlit run app.py

• Keep hr_employee_raw.csv beside app.py.

• Demonstrate department, employee status, work mode, and job-level filters.

Instructor note: because this is a challenge-based session, provide the CSV and business questions first. Reveal code hints progressively. The dashboard and starter script are reference materials for checking work, not the first thing students should copy.